

1. Kinematics: Exercises

Exercises 1

A particle moves according to the equation $x = 10t^2$, where x is in meters and t is in seconds.

- Find the average velocity for the time interval from 2.00 s to 3.00 s.
- Find the average acceleration for the time interval from 2.00 to 2.5 s.

Exercises 2

A position-time graph for a particle moving along the x axis is shown in Figure(1).

- Find the average velocity in the time interval $t = 1.50$ s to $t = 4.00$ s.
- Determine the instantaneous velocity at $t = 2.00$ s by measuring the slope of the tangent line shown in the graph.
- At what value of t is the velocity zero?

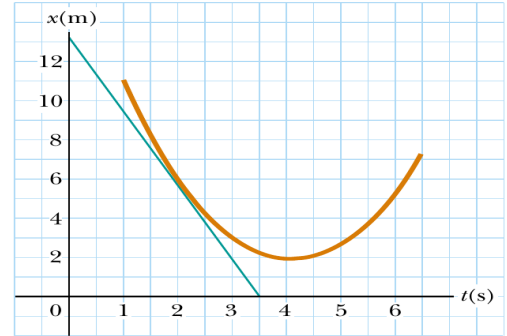


Fig. 1

Exercises 3

Figure (2) corresponds to the acceleration versus time graph of a mobile (A) moving parallel to the x -axis. At $t = 0$ s, $x_A(0) = 0$ m and $V_A(0) = 2$ m/s.

- Plot the velocity graph $V_A(t)$ of the mobile (A). We choose : $1cm \rightarrow 2s$ and $1cm \rightarrow 0.5m/s$.
- Find the mobile position at $t = 8$ s.
- An other mobile B crosses the axis origin at $t = 0s$, moving with a constant velocity, reaches the same position of (A) at $t = 8$ s. Find the velocity of the mobile B? At what time (B) would meet (A) again?
- Which one of the mobile (A) or (B) is at the first position at $t = 20$ s?

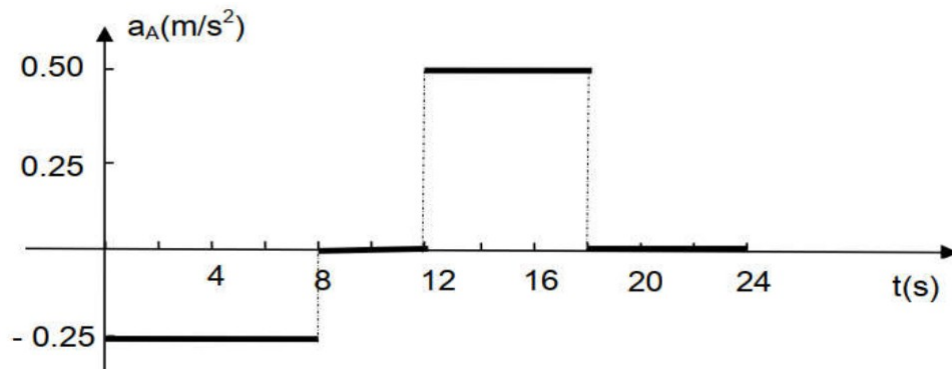


Fig. 2

Exercises 4

A golf ball is hit off a tee at the edge of a cliff. Its x and y coordinates as functions of time are given by the following expressions:

$$x = 18.0t(m), \quad y = 4.00t - 4.90t^2(m)$$

- Determine the trajectory equation. What is the type of the motion
- Write a vector expression for the ball's position as a function of time, using the unit vectors $\hat{i}$ and $\hat{j}$.
- obtain the expressions for the velocity vector $\mathbf{v}$ and the acceleration vector $\mathbf{a}$ as a function of time.

4) Use unit-vector notation to write expressions for the position, the velocity, and the acceleration of the golf ball, all at $t = 3.00 \text{ s}$.

Exercises 5

A gun fires a bullet with a velocity of 200 m/s at an angle of 40° with the ground. Find the velocity and position of the bullet after 20 s . Also find the range, maximum height and the time required for the bullet to return to ground.

Exercises 6

A particle is moving in a circle according to the law $\theta(t) = 3t^2 + 2t$, where θ is measured in radians and t in seconds. Calculate the angular velocity and angular acceleration after 4 s .

Exercises 7

A body, initially at rest ($\theta = 0$ and $\omega = 0$ at $t = 0$) is accelerated in a circular path of radius 1.3 m according to the equation $\alpha(t) = 12t^2 - 48t + 16$. Find the angular position and angular velocity of the body as functions of time, and the tangential and centripetal components of its acceleration.

Exercises 8

An object is moving in the xy plane with coordinates varying with time according to the equations

$$x = -5.00 \sin(\omega t) \text{ (m)}; \quad y = 4.00 - 5.00 \cos(\omega t) \text{ (m)}$$

where t is in seconds and ω has units of s^{-1} .

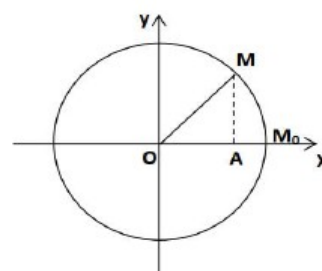
1. Determine the trajectory equation. What is the type of the motion?
2. Write expressions for the position vector, velocity vector, and the acceleration vector at any time $t > 0$ in the Cartesian system of coordinate.
3. Write expressions for the velocity and acceleration vectors at any time $t > 0$ in the polar and intrinsic system of coordinate.

Exercises 9

The circular trajectory of a mobile (M) corresponds to the figure by using a $(1/100)$ scale. At $t_0 = 0 \text{ s}$ M position is M_0 and the angular velocity is:

$$\omega(t) = \begin{cases} \alpha t & \text{for } t \leq 2 \text{ s} \\ \pi & \text{for } t \geq 2 \text{ s} \end{cases}$$

where ω has unit rad/s , t in s and α is a constant.



1. Find the value and unit of α . What is the corresponding physical meaning. Find the radius $r(t)$ and the angular coordinate versus time.
2. Give the mobile curved abscissa $M(t = 2.5 \text{ s})$.
3. Find expressions of the radial v_r and the transversal v_θ component of the vector velocity as function of time.
4. Find as function of time, the intrinsic component a_t and a_n of the acceleration vector. Give the the radial a_r and transversal a_θ components of the acceleration vector. Precise the motion phases.
5. At $t = 2 \text{ s}$, a mobile (A) starts a motion along the (Ox) axis. The coordinate x_A is given by the Cartesian component x_M of the mobile (M).
 - a) Give the equation of motion $x_A(t)$ and the motion's period T .
 - c) Find the expression of the velocity and the acceleration of (A) in term of time.